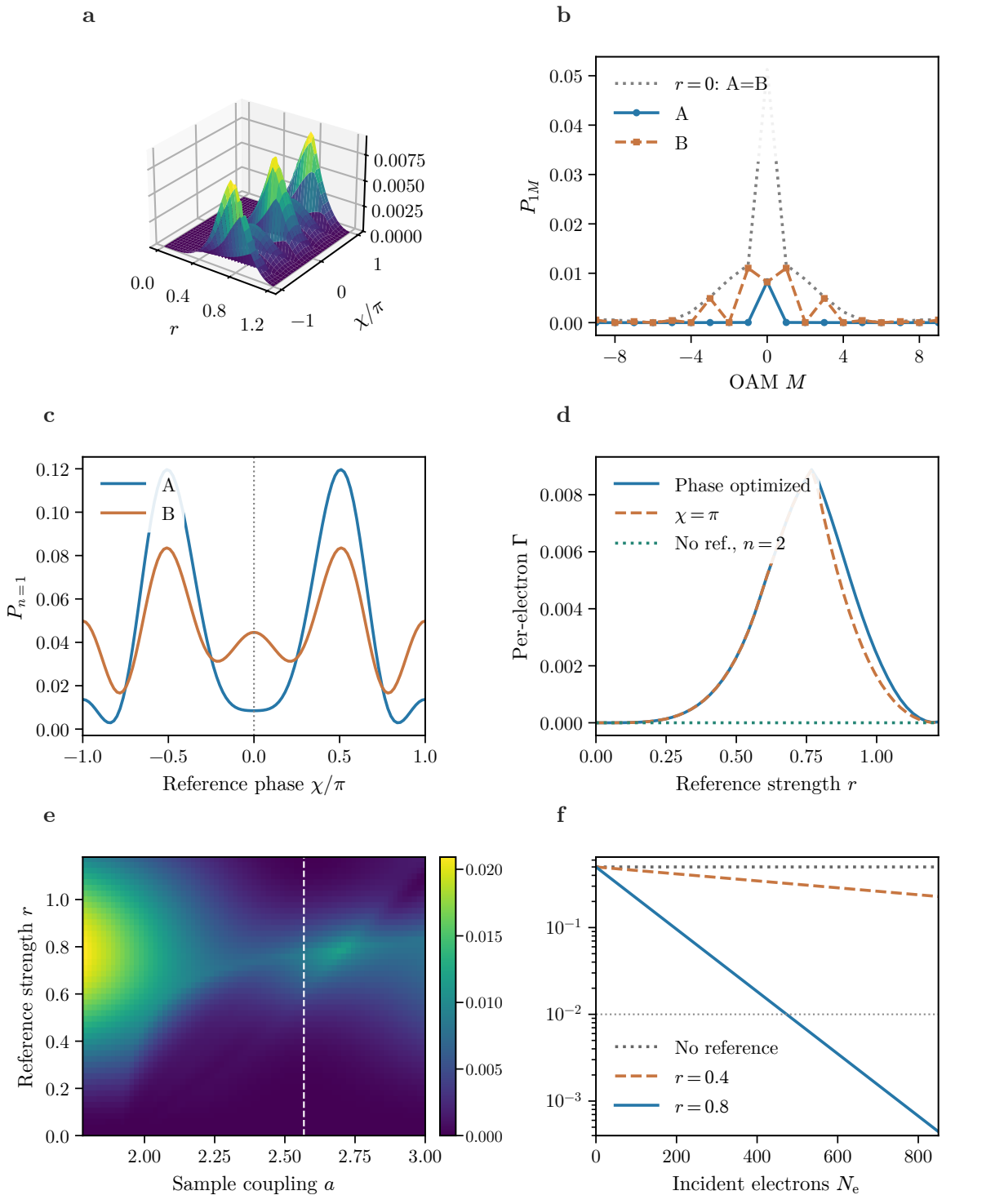

a
0.0075
0.0050
0.0025
0.0000
0.0
0.4
0.8
1.2
r
−1
0
1
χ/π
b
r = 0: A=B
A
B
0.05
0.04
0.03
0.02
0.01
0.00
P1M
−8
−4
0
4
8
OAM M
c
A
B
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Pn=1
−1.0
−0.5
0.0
0.5
1.0
Reference phase χ/π
d
Phase optimized
χ = π
No ref., n = 2
0.008
0.006
0.004
0.002
0.000
Per-electron Γ
0.00
0.25
0.50
0.75
1.00
Reference strength r
e
1.0
0.8
0.6
0.4
0.2
0.0
Reference strength r
2.00
2.25
2.50
2.75
3.00
Sample coupling a
0.020
0.015
0.010
0.005
0.000
f
10−1
10−2
10−3
No reference
r = 0.4
r = 0.8
0
200
400
600
800
Incident electrons Ne